Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/13/2005
PI Name:	Su, Ching-Hua Ph.D.		
Project Title:	Structural Fluctuations and Thermophysical Properties of Molten II-VI Compounds		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Marshall Space Flight Center		
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City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	10/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Feth, S. (University of Alabama/NASA/MSFC) Lehoczky, S. (NASA MSFC) Mook, H. (Oak Ridge National Library) Scripa, R. (University of Alabama at Birmingham) Zhu, S. (AMSAM-TMD-LW)		
Grant/Contract No.:	Intramural		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To support the two microgravity materials science flight experiments entitled "Crystal Growth of Selected II-VI Semiconducting Alloys by Directional Solidification" and "Growth of Solid Solution Single Crystals", we propose to perform structural and thermophysical property measurements on the II-VI melts which include the HgTe binary as well as the pseudobinaries of HgCdTe and HgZnTe. The measurements include neutron scattering, viscosity, electrical conductivity, thermal diffusivity, density and volumetric expansion coefficient. The measurements will be conducted under two sets of experimental conditions. (1) Measurements as functions of time near the liquidus point, including at temperatures below the liquidus temperature on the supercooled liquid, will provide basic information on heterophase fluctuation at the vicinity of the phase transition. (2) Measurements as functions of temperature from the liquidus to higher temperatures will enhance the fundamental understanding of the structural transition as well as the semiconductor-metal transition, which typically covers a wide temperature range. To support the two microgravity materials science flight experiments entitled “Crystal Growth of Selected II-VI Semiconducting Alloys by Directional Solidification” and “Growth of Solid Solution Single Crystals”, we propose to perform structural and thermophysical property measurements on the II-VI melts which include the HgTe binary as well as the pseudobinaries of HgCdTe and HgZnTe. The measurements include neutron scattering, viscosity, electrical conductivity, thermal diffusivity, density and volumetric expansion coefficient. The measurements will be conducted under two sets of experimental conditions. (1) Measurements as functions of time near the liquidus point, including at temperatures below the liquidus temperature on the supercooled liquid, will provide basic information on heterophase fluctuation at the vicinity of the phase transition. (2) Measurements as functions of temperature from the liquidus to higher temperatures will enhance the fundamental understanding of the structural transition as well as the semiconductor-metal transition, which typically covers a wide temperature range. High quality single crystals of compound semiconductors are needed for various elctro-optical applications. At present, the majority of semiconducting crystals are grown from the melt. Consequently, the pre-crystallization phenomena in the liquid phase are critical because the state and structure of the melt affects the structure and properties of the crystals formed from it. A fundamental understanding of molecular structural dynamics in the melt will be helpful in understanding the relations between the interfacial flows and molecular attachment mechanisms, including those generating lattice defects. Therefore, an in-depth study on the time-dependent structural dynamic processes taking place in the vicinity of the solid-liquid phase transition as well as the in-situ structural analysis of the alloy homogenization process in the melts is needed to understand and to improve the crystal growth processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Neutron Scattering Experiments: 1. A neutron diffraction experiment was performed for liquid Tellurium at Argonne National Laboratior, Intense Pulsed Neutron Source (IPNS), using the Glass Liquid, and Amorphous Materials Diffractometer (GLAD), from December 6-14, 2004. The diffraction spectra were taken at sample temperatures ranged from melting point 450 to 750 degrees C with the addition of a supercooled temperature of 430 degrees C. The data are being analyzed to examine the possible change of liquid structure from two- to three-fold domain at around 600 degrees C. 2. A neutron diffraction experiment on HgZnTe and Te melts was performed at IPNS Argonne National Laboratory from August 20 to 26, 2003. The experiment was awarded to study the phase structure and transformation in molten HgZnTe and Te to suppport the ground-based project. The preliminary results showed a time-dependent drift of the scattering spectra even after 10 hours at constant temperature.
Bibliography Type:	Description: (Last Updated: 02/18/2025)
Articles in Peer-reviewed Journals	Li,C, Ban H, Scripa RN, Su CH, Lehoczky SL. "Transient torque method: A fast and nonintrusive technique to simultaneously determine viscosity and electrical conductivity of semiconducting and metallic melts." Review of Scientific Instruments, 2004 Sep;75(9):2810-6. http://dx.doi.org/10.1063/1.1781380 , Sep-2004
Articles in Peer-reviewed Journals	Barlow DA, Baird JK, Su CH. "A theory of the von Weimarn rules governing the average size of crystals precipitated from a supersaturated solution." Journal of Crystal Growth. 2004 Mar 15;264(1-3):417-23. http://dx.doi.org/10.1016/j.jcrysgro.2003.12.047 , Mar-2004
Papers from Meeting Proceedings	Ban H, Lin B, Li C, Scripa R, Su CH, Lehoczky SL. "Torque Transient of Magnetically Driven Flow for Viscosity Measurements." 2004 Heat Transfer/Fluids Engineering Summer Conference (HT/FED'04), Charlotte, NC, July 11-15, 2004. 2004 Proceedings of the ASME Heat Transfer/Fluids Engineering Summer Conference, 2004. HT-FED2004-56715. ISBN #: 0791846903 , Jul-2004
Presentation	Lin, B.; Zhu, S.; Ban, H.; Li, C; Scripa, R.; Su, C.H.; Lehoczky, S.L. "Modified Laser Flash Method for Thermal Property Measurements and the Influence of Heat Convection" 2003 ASME International Mechanical Engineering Congress and R&D Expo Washington D.C. Nov-2003
Presentation	Ban, H.; Lin, B.; Li, C.; Scripa, R.; Su, C.H.; Lehoczky, S.L. "Torque Transient of Magnetically Driven Flow for Viscosity Measurements" 2004 Heat Transfer/Fluids E ngineering Summer Conference Charlotte North Carolina, July 11-15, 2004 Jul-2004
Presentation	Li, C.; Ban, H.; Lin, B.; Scripa, R.N.; Su, C.H.; Lehoczky S.L. "Temperature Dependence of Density, Viscosity and Electrical Conductivity for Hg-Based II-VI Semiconductor Melts" 14th International Conference on Crystal Growth (ICCG-14), Grenoble, France Aug-2004

Presentation	Ban, H.; Li, C.; Lin B.; Emoto, K.; Scripa R.N.; Su, C.H.; Lehoczky, S.L. "Thermal Diffusivity for III-VI Semiconductor Melts at Different Temperatures" 14th International Conference on Crystal Growth (ICCG-14), Grenoble, France Jan-2004
Presentation	Li, C.; Su, C.H.; Lehoczky, S.L.; Scripa, R.N.; Ban, H.; Lin, B. "Thermophysical Properties of Selected II-VI Semiconducting Melts" Fourth International Conference on Solidification and Gravity (SG04), Miskolc-Lillafured, Hungary Sep-2004
Presentation	Li, C.; Lehoczky, S.L.; Su, C.H.; Scripa, R.N. "Electrical Conductivity of HgTe at High Temperature" 2004 Materials Research Society (MRS) Fall Meeting, Boston MA Nov-2004